



James Evans <jamwevan@umich.edu>

to Oyakhi, CHONG, Tolulope ▾

9:15 PM (18 minutes ago)



Hello Dr. Ibhagui,

I spent some more time thinking about the areas of concern in the abstract. One sentence in particular still raises questions: *"In contrast, the COVID-19 period exhibits a breakdown of long-run cointegration despite elevated volatility, driven by the emergence of a new, distinct global stochastic trend which increases the number of stochastic trends and erodes long-run linkages, reflecting greater currency-specific autonomy."*

It seems that through this phrasing we are referencing Proposition 2: the keyword here being volatility. But, Proposition 2 is not making a claim about cointegration breaking down: instead, it describes how correlations rise through two channels: (i) the variance channel and (ii) the loading-alignment (autonomy compression) channel. Furthermore, how can we connect Proposition 2 with a breakdown of long-run cointegration when we are operating under the assumption that the $\text{rank}(\Lambda_r)$ is unchanged? These two reasons lead me to believe that a breakdown in long-run cointegration cannot be counteracted by elevated volatility. I tried reconciling this discrepancy with the point brought up in your previous email:

"If Cointegration, then Strong Long Horizon Returns Correlations with High magnitude $\rightarrow 1$. This statement is the same as saying the Contrapositive: If Weak Long Horizon Returns Correlation $\rightarrow 1$, then No Cointegration."

But, after rereading Proposition 2, I was unable to do so. Proposition 2 states:

"If $\text{rank}(\Lambda_r)$ remains unchanged in crisis, but (i) $\sigma^2_{g_{k,r}}$ rises for one or more factors, and (ii) $|\lambda_{ik,r}|$ converge across currencies (autonomy compression), then long-run correlations increase."

The problem is that long-run correlations increasing has nothing to do about the strength of the correlation. Correlation can go from .1 to .15, exhibit an increase, and still be considered weak. I propose deleting the phrase "despite elevated volatility" to fix the sentence. This edit will make the connection to Proposition 3 even more clear and make the statement logically coherent with the ideas presented in the paper.

This other sentence also still raises questions: *"Our findings confirm the **existence of cointegration** among major currencies during the Global Financial Crisis, **consistent with reduced currency autonomy driven by common global risk factors.**"*

This sentence seems to be referencing the loading-alignment channel. However, Proposition 2 speaks to correlations rather than cointegration, while this phrasing appears to suggest a link between reduced currency autonomy and cointegration. For the same reasons as before, I am unable to reconcile this discrepancy with the point you brought up in email with me earlier.

Also, can you explain to me how we plan on incorporating the data in the spreadsheet you sent to us earlier?

You said in your earlier email: "The Covid and Dot Com are more consistent with the contrapositive: weak correlation, no cointegration (0/4). The GFC less so, but you can argue GFC is also consistent because a) correlation is also weak and b) although you have cointegration, it is just 1 out of 4 max."

But I fail to see how this data directly supports any of our propositions empirically. This is the only idea I managed to come up with:

We use the methodology—first applying unit root tests to confirm that the series are $I(1)$, and then conducting system tests to identify long-run relations—to establish that the cointegration rank was 0 during the crisis. If we assume that this cointegration rank remained unchanged throughout the period, then we can conclude that $\text{rank}(\Lambda_r)$ is unchanged—satisfying the first condition of Proposition 2.

The second condition also seems straightforward: σ^2 rising should be a natural market response to crises, including this one. With both suppositions satisfied, we can reasonably apply the variance channel. If we plot correlations over time and run a least-squares regression, we may be able to demonstrate an increase in correlation consistent with the result in Proposition 2.

This would be particularly meaningful. The pandemic represents a "novel phenomenon absent in prior crises and largely unexamined in cointegration literature." It challenges the standard crisis-contagion narrative by showing that global crises do not always uniformly increase FX integration. Being able to explain this phenomenon through the proposition proposed in our paper and provide empirical evidence on top of the proof we already have would advance the existing literature on this topic in a compelling way.

Warm regards,

James Evans

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Dear James, and Chong read carefully too, as I try to settle all confusions with this lengthy explanation:

Generally, there is the common stochastic trends [common global factors that all currencies load on] and there is the idiosyncratic stochastic trend. When you combine both, you have the full stochastic trends. Contingent rank $N = \text{number of common stochastic trends}$, where the number of common stochastic trends is the rank of α, Γ, L , the common stochastic trends. Here, there is always an implicit assumption that there are no idiosyncratic stochastic trends because once they arise, they can kill and will kill cointegration even when there were initially common stochastic trends. How? Because now, they make some variables to drift away and do their own thing, destroying the common trend that everyone should have that would lead to cointegration. Antonian then starts developing. One symptom is that they no more respond similarly to the same common shocks; the lambdas can be different.

Autonomy generally arises from people doing their own thing, i.e. idiosyncratic. One direct implication of this is that they do not respond similarly to whatever common stochastic trend exists, because perhaps they are responding in special ways to their own idiosyncratic stochastic trends.

If rank of $\mathbf{c}_j$ does not increase, it simply means that there wasn't an additional common stochastic trend that all the currencies would respond to as fresh, perhaps similarly. This doesn't mean that there cannot be an emergence of some idiosyncratic stochastic trend. When such a trend emerges when the common stochastic trend hasn't changed, this could cause autonomy among the currencies because they wouldn't respond to the existing common factors in the same way that they should. Proposition 2 makes clear that if they are responding similarly to the common stochastic trend factors (and they only do this when there is no sudden idiosyncratic stochastic trend that would make the currencies respond in their own different ways and lead to greater autonomy), then long horizon returns correlation would be closer to 1.

After reading the first two paragraphs, take some time to read this paragraph you quoted again

^{***}In contrast, the COVID-19 period exhibits a breakdown of long-run cointegration despite elevated volatility, driven by the emergence of a new, distinct global stochastic trend which increases the number of stochastic trends and erodes long-run linkages, reflecting greater currency-specific autonomy.

As far as the word "despite" goes, perhaps you missed Chong's email yesterday or didn't carefully read it. That matter has been discussed and settled; I mean the word "despite". Cointegration is not being countered by increased or elevated volatility. I had written a short but direct response to this exact point to Chong. I suggest you read that carefully, with my leg press and research example. If you disagree with the suggestion given, you should contest the suggestion. But yes, that has been discussed yesterday.

Proposition 2 contains in the possibility of a contagion breakdown. A section of it says if the long horizon returns correlations do not go to 1, those lamdas shouldn't lamda similarly. If those lamdas don't lamda similarly, then by definition someone is having an autonomy that is perhaps caused by something idiosyncratic, so no more responding similarly to common factors. Contagion weaknesses, as in the case of GFC, or breaks down entirely as in the case of Covid-19. This is why it directly follows that, perhaps a new idiosyncratic stochastic factor arose during Covid-19 that destroyed the commonality that normally should have existed. Proposition 2 gives a possible guide to what possibly happened. So, yes, Proposition 2 has in it some condition about a contagion breakdown, even if it is not only about contagion breakdown. You only need to read the Contrapositive and then use the earlier parts of this email to reach a conclusion on your Contrapositive and also relate it to a breakdown of contagion.

Now, direct responses to your questions:

1. How can we connect Proposition 2 with a breakdown of long-run cointegration when we are operating under the assumption that the rank(Λ) is unchanged?

to think(1), that is the common stochastic global factor trend, does not change, but then if you suddenly have autonomy due to the variables now responding similarly anymore to the common stochastic global factors, then a likely distractor, an idiosyncratic stochastic trend which was initially non-existent, has risen up, caused some of the variables to "do their own thing", and essentially killed what would others have been a similar response to the existing (and unchanged) common global stochastic trend. This is one way that cointegration can break down when κ is unchanged. Of course, there are other ways, one of which is that there is no common stochastic trend at all (in which case all the variables are individually stationary) or all of them are non-stationary and not cointegrated because all trends are idiosyncratic stochastic trends and each has variable its own idiosyncratic stochastic trends. There is still one or two, but we're focusing on your question.

2. These two reasons lead me to believe that a breakdown in long-run cointegration cannot be counteracted by elevated volatility.

I think I have addressed this already. The paper does not say breakdown of cointegration is because of or lack of volatility

I hope you can now see that point (ii) implies cointegration, *ceteris paribus*. Autonomy compression triggers cointegration

4. The problem is that long-run correlations increasing has nothing to do about the strength of the correlation. Correlation can go from .1 to .15, exhibit an increase, and still be considered weak. I propose deleting the phrase "despite elevated volatility" to fix the sentence. This edit will make the connection to Proposition 3 even more clear and make the statement logically coherent with the ideas presented in the paper.

[illegible]

5. This sentence also still raises questions: "Our findings confirm the **existence of cointegration** among major currencies during the Global Financial Crisis, **consistent with reduced currency autonomy driven by common global risk factors**." This sentence seems to be referencing the loading-alignment channel. However, Proposition 2 speaks to correlations rather than cointegration, while this phrasing appears to suggest a link between reduced currency autonomy and cointegration. For the same reasons as before, I am unable to reconcile this discrepancy with the point you brought up in email with me earlier.

Once again, proposition 2 in part, connects correlation and contagion. When you have contagion, several things have happened: the means of the common denominator that has now tied them together and caused them to co-integrate. The second, of course and as I have tried to explain, is that there is no idiosyncratic stochastic trend that would completely kill the contagion. The other possible reason for fewer contagion could be that there are more common stochastic factors. That too can kill contagion. Too much cooking spoils the broth. This is the reason why rank of the common stochastic trends q is always less than or less than N . If it is so many that it equals N , contagion is gone. You could argue that this could be a reason why we didn't have contagion in Covid-19 because the common factors increased to N . But then, it was already stated that the rank of q didn't change, so that increase is ruled out. Which then follows that the lack of contagion in that period most likely stems from the

5. Also, can you explain to me how we plan on incorporating the data in the spreadsheet you sent to us earlier?

I sent you the data for two reasons. First is to make you double check the results we have; second, for you to creatively think about what additional beautification you can come up with in the data that will fit well in the paper. For example, charts and maybe some other creative stuff based on you and Chong's own intuition. You and Chong to use your creativity to expand sections of the paper, and this includes figuring out what additional thing to do with the data that would improve the paper. Lots of ideas on google scholar.

6. You said in your earlier email: "The Covid and Dot Com are more consistent with the contrapositive: weak correlation, no cointegration (0/4). The GFC less so, but you can argue GFC is also consistent because a) correlation is also weak and b) although you have cointegration, it is just 1 out of 4 max."

Please think carefully about this question or comment you made. Among the set of rich people, there are richer people. Does it mean that the richer less rich people are not rich? No, it does not. If you rank the three crisis periods, the three of them have similar correlations on average. The correlations are nowhere near around 1, on average. Two have them have zero conintegration out of 4, one of them has 1 evidence of conintegration out of 4. Therefore, those two are "worse off" than the other one. Or is your point that 1 conintegration = 0 conintegration? In words, 1 is approximately 0? That's what you're implying if you try to equate the consistency of these three. That Covid and Dot Com are more consistent does not mean that GFC does not share some similar ingredients with them. It just means Covid and Dot Com are more. A and B are impressive. B is more impressive than A. So, A is not impressive that won't work.

7. But I fail to see how this data directly supports any of our propositions especially (1), the only ideal I managed to come up with: the methodology – first applying unit root tests to confirm that the series are $I(1)$, and then conducting system tests to identify long-run relations – to establish that the cointegration rank was 0 during the crisis. If we assume that this cointegration rank remained unchanged throughout the period, then we can conclude that rank(A) is unchanged – satisfying the first condition of Proposition 2. The second condition also requires the cointegration rank to be 0 during the pandemic period. We can also market responses to this condition by saying that the pandemic period is a “novel phenomenon absent in prior (A) and largely unexamined in cointegration literature”⁷ It challenges the standard cointegration narrative by showing that rank is not always 1, and that the cointegration rank can be 0. The FX integration phenomenon is not only a novel phenomenon in the paper and the empirical literature, but also a novel phenomenon in the cointegration literature. This is a compelling way.

I think or believe that the previous lengthy explanations have addressed your point 7. You can lose cointegration even when $\text{rank}(\Lambda)$ is unchanged. I would suggestion revisiting the earlier paragraphs of this email

Best

Oyakhi



Oyakhi Ibhagui

to Tolulope, CHONG, me ▾

Tue, Sep 2, 10:21 PM (2 days ago)



Dear Chong,

1) Volatility rose during Covid (i.e. a real-life market outcome); 2) After performing cointegration analysis, we found no cointegration; 3) but, did long horizon correlations rise? Maybe,.. but those correlations look small.

1) and 2) appear like things we can defend. So, we need to look for the right word to join them.

Instead of *despite*, maybe we try "*amid*". Or any other creative word. Two things can be right at the same time without one implying the other: a) I'm a finance researcher at BAUM TenPers and b) I lift 900 pounds leg press at the gym ...Two things that are true yet unconnected, at least not directly.

Regards

Oyakhi

✓ In short:

"Common" trends are shared, but if there are as many as N , then every currency is explained by a distinct mix of them, leaving no redundancy to generate cointegration. True independence comes either from idiosyncratic trends (ruled out by assumption) or from having too many common factors (so the overlap vanishes).



you said: We want the number of common stochastic trends to be limited because cointegration requires overlap — currencies must share trends. If every currency has its own independent trend, they all drift apart and you lose cointegration entirely. But how can each currency have its own independent trend when we are talking about common stochastic trends?





Oyakhi Ibhagui

to me, CHONG, Tolulope ▾

8:40 PM (1 hour ago)



Hello James

It's good to know it helped.

Once you guys finish this and then we do a zoom presentation where you both prepare and do a presentation of the same paper, it will be super clear — you will see all the strengthens and all the weaknesses the paper has.

So, the plan is for us to receive the final polished write up and then corrections will be suggested and then you both will present the work on zoom.

Lastly, if you are thinking of proposition 2, have this in mind: If (A and B) then C. The contrapositive is: If not C, then not (A and B)...And then De Morgan (or whatever you math guys call it) would mean If not C, then not A or not B. If we know that not B is likely to false (for example, say B is volatility increase, then not B during a crisis is very likely false as one of the hallmarks of crisis is some serious volatility), then our contraposition just shrinks to If not C, then not A. With this framework, replacing C with long horizon correlations and A with similar loadings of those common factors on the currencies, things become clearer.

I was contemplating adding the below scheme to the appendix, but I will leave it to you both to think it through, determine if it is important to add to the appendix or if it is not.

- Start: **If (A and B) → C**
(A = similar loadings; B = volatility rises; C = long-horizon correlations → 1).
- Contrapositive: **If not C → not (A and B)**.
- By De Morgan (or what's the name?): **If not C → (not A or not B)**.
- In crisis settings, **not B is implausible** (volatility almost always rises).
So effectively: **If not C → not A**.

End: If long-horizon correlations don't near 1 (i.e. low), currencies' factor loadings are not similar — i.e., autonomy and idiosyncratic trends possibly dominate, kill cointegration.

Regards

Oyakhi